empirical theories, 155–156
empty space
  atoms as, 34, 73
  energy of, 256–257
  entropy and, 278
  no particles in, 260–261
  quantum vacuum in, 256–257, 259–261
  quantum version of, 302
  as stationary, 260
energy, 173, 184, 253, 256–257, 281–282, 291. *See also* vacuum energy
entangled superposition, 114–116
entanglement. *See also* Bell's theorem on entanglement
  in action, 94–95
  to define distances, 276–277
  degrees of freedom in, 263, 283–284
  description of, 37–38, 91
  in different regions of space, 261–265
  Einstein and, 31
  electrons and, 92, 120–122
  entropy and, 160
  with environment, 118–119
  EPR paper on, 96–102, 109
  examples of, 91–92
  in GRW theory, 182–183
  momentum and, 92
  nonlocal nature of, 178
  no-signaling theorem, 97–99
  in quantum field theory, 249
  quantum states and, 118, 261–262
  quantum systems and, 159–160
  Schrödinger's equation and, 38
  two-qubit system, 95–96
  vacuum energy and, 262–265
entanglement entropy, 160, 277, 283
entropic arrow of time, 158–159
entropy
  area and, 285
  arrow of time, 158–159
  atoms and, 158, 160, 276, 297–298
  Bekenstein-Hawking, 297–299
  of black holes, 297–299
  Boltzmann formula for, 158–163
  in closed systems, 158
  cutoffs, 278
  of empty space, 278
  entanglement, 160
  event horizon as similar to, 297
  limits on, 300
  low, 159
  from objective to subjective feature, 161–162